Zhang Xueba

Phone: 138-0000-0000 | Email: zhangxueba2027@163.com | Degree: Master's
Hometown: Dongcheng, Beijing | Political Affiliation: CPC Member | Gender: Male

Education

University of Electronic Science and Technology of China (2023.09–Present) (985, Double First-Class) Master's | School of Information and Communication Engineering | Electronic Information Engineering (A+) | GPA: 90/100 | Top 5%

University of Electronic Science and Technology of China (2019.09–2023.06) (985, Double First-Class) Bachelor's | School of Information and Communication Engineering | Information and Communication Engineering (A+) | GPA: 90/100 | Top 5% (recommended for postgraduate study)

Research Interests: Multimodal Large Models, Reinforcement Learning, Agent Development

Publications

- **SAM-Adapter: Lightweight Domain Adaptation for Segment Anything Model via Prompt Tuning** (First Author, CCF-A, ICCV-25)
- **Dual-Granularity Prompt Learning for CLIP-based Incremental Few-Shot Learning** (Second Student Author, CCF-B, ACM MM-24)
- **Multi-Scale Fourier Enhanced Vision Transformer for Remote Sensing Change Detection** (Corresponding Author, CAS Q1, ISPRS PRS (under review))
- **Hypergraph Neural Networks with Cross-Modal Contrastive Learning for Recommendation** (First Author, CCF-B, DASFAA-24)

Internship Experience

xx Big Tech Company | Algorithm Engineer | xx Department **2025.05–2025.08**

- **Main Work:** Participated in upgrading the large-scale recall pipeline of the company's main-search "Guess You Like" feature, leading the deployment of **multi-interest representation and Generative Retrieval (GR)**. Proposed a **hierarchical contrastive learning and vector caching mechanism** to replace the traditional inverted index, significantly improving recall diversity and relevance during the Double 11 stress test; online CTR increased by **3.2%**.
- **Technical Contributions:**
  (1) To mitigate **noise interference and long-tail distribution in user behavior sequences**, built a Transformer-based multi-interest extraction framework. Introduced an **adaptive Noise Contrastive Estimation (NCE)** loss to dynamically adjust negative-sample weights, decoupling users' implicit interests into 8 orthogonal vectors; built a billion-scale item index with **Faiss**, and offline Recall@100 improved by **5.8%**.
  (2) To address the **poor generalization of traditional recall models on unexposed items**, implemented a **Diffusion-based Generative Retrieval** model. Abandoned traditional Item IDs and instead encoded items as semantic IDs, training a DiT model to generate target semantic-ID sequences directly from the Query; designed a **Curriculum Learning** training strategy that learns the high-frequency item distribution first and hard samples later; new-item recall rate improved by **12.5%**.
  (3) To resolve the engineering bottleneck of **high online inference latency and GPU memory usage**, developed an **adaptive KV Cache quantization and compression module**. Applied INT8 quantization to the generative retrieval model with TensorRT-LLM and optimized memory fragmentation with PagedAttention; deployed on an A10 GPU cluster, single inference latency was kept within **15ms** and QPS increased by 40%; launched after passing A/B testing.

xx Big Tech Company | Algorithm Researcher | xx Department **2025.10–Present**

- **Main Work:** Responsible for R&D of **Diffusion Transformer (DiT)-based large video generation models**, focusing on improving **motion consistency** and **instruction following** in short-video generation. Participated in building the underlying generation engine of CapCut's "Image-to-Video" feature, optimizing inference efficiency on consumer GPUs to support real-time 1080P/24fps video generation.
- **Technical Contributions:**
  (1) To address **temporal flickering and object deformation in video generation**, designed a **Spatial-Temporal Joint Attention** mechanism. Introduced 3D convolutions in the VAE encoding stage to extract spatio-temporal features, replacing the traditional separate spatial/temporal modules and significantly improving inter-frame continuity; on WebVid-10M, FVD dropped to **128.5**, a **15.3%** improvement over the baseline.
  (2) To address **poor understanding of complex Chinese Prompts**, built a million-scale video-text alignment dataset. Proposed a **hierarchical Caption annotation Pipeline**: first used GPT-4V to produce fine-grained descriptions (objects, actions, lighting) of key video frames, then trained **Qwen1.5-14B** as a Reward Model (RM) for quality scoring and filtering. Performed full-parameter fine-tuning of Stable Video Diffusion on this data; CLIP Score rose to **0.42** and instruction-following accuracy improved by **22%**.
  (3) To break the **GPU memory wall of high-resolution video inference**, implemented a **cascaded Latent Consistency Distillation (LCD)** algorithm, compressing the original 50-step sampling to **4 steps**, combined with TensorRT FP16 quantization and Kernel Fusion; on an A10 GPU, single-frame inference latency dropped from 3.2s to **0.4s** and memory usage decreased by 60%, supporting the gray-release launch of the app's "One-Click Video" feature.

Project & Research Experience

xxx Smart Logistics Park Scheduling System | Research Lead / One of the Technical Leads | University– 2024.11–2025.05
Enterprise Cooperation with a Large Logistics Company

Project Overview: In partnership with a leading local logistics company in Sichuan, built a smart logistics park scheduling system integrating **lightweight visual detection models and operations-research optimization algorithms**, achieving real-time tracking and dynamic path planning for all park elements (unmanned forklifts, sorters, pallets, shelves). By building a **large-scale logistics-scene trajectory dataset**, trained a dedicated scheduling model and built a Multi-Agent collaborative decision system; the core scheduling module has been integrated into the company's "Shuyuntong" WMS for trial operation.

Technical Contributions: (1) To address **severe target occlusion and scarce annotation data under park surveillance views**, adopted a **semi-supervised learning** framework combined with **active learning** to select highly uncertain samples for manual annotation; used **CutMix** and **Mosaic** augmentation to simulate cargo stacking occlusion; used **ControlNet** with edge-detection maps to generate synthetic data of specific shelf layouts, expanding the training set by 30%; used **StyleGAN3** for cross-camera style transfer to improve cross-domain generalization.

(2) To address **difficult fine-grained recognition caused by many unmanned forklift models**, adopted a **Cascade Mask R-CNN** multi-stage detection architecture that cascades three detection heads with progressively higher IoU thresholds, achieving precise recognition from generic industrial vehicles to specific forklift models; mAP50 improved by 4.1%. For **missed detection of dense small targets from high overhead views**, introduced **Deformable Convolution (DCN)** to enhance feature extraction and used **sliding-window slicing inference**; for **limited edge compute**, applied INT8 quantization to the detection model and optimized the inference engine with **TensorRT-LLM**, while cascading the tracker with **Kalman filtering and the Hungarian algorithm**, ultimately achieving stable 45FPS detection and tracking on Jetson Orin.

(3) For **low path-planning efficiency under complex road conditions** and decisions that do not require frame-by-frame inference, used an **improved A*** algorithm with **time-window conflict detection** for global path planning, and used **Deep Reinforcement Learning (PPO)** to train a local obstacle-avoidance policy network; the PPO model was trained on 5,000+ hours of collected real park trajectories, 20k+ manually simulated congestion scenarios and 8,000+ scheduling-rule records; task completion rate rose to 96.5%.

(4) To build a **vertical-domain intelligent decision assistant for logistics scheduling**, built a **large-scale logistics knowledge graph** and **natural-language interaction system**. Extracted order, inventory and capacity data from ERP/WMS via ETL tools to build a knowledge graph with 100k+ entity nodes, stored in Neo4j; fine-tuned **Qwen2.5-32B** with LoRA on 5,000+ historical scheduling logs and emergency plans as the decision base model; finally built a **Multi-Agent collaborative scheduling system** using the **ReAct** agent pattern with **Chain-of-Thought (CoT)** prompting to optimize decision logic; compared with the traditional rule engine, response speed in abnormal-event handling improved by 12

Honors

Scholarships: National Scholarship (2022, 2021, 2024), National Encouragement Scholarship (2020), Outstanding Student Scholarship (2020, 2021, 2022, 2023).

Competitions: xxx Algorithm Challenge — National First Prize; Graduate Artificial Intelligence Innovation Competition — National First Prize; National College Student Information Competition — First Prize; Lanqiao Cup Information Competition — First Prize;

Professional Skills

Algorithms: Solid theoretical foundation in **Large Multimodal Models (LMMs)** and DiT architectures; proficient in **object detection**, image segmentation (SAM) and multimodal alignment; deep understanding of the full **RLHF** pipeline, familiar with the principles and code implementation of preference optimization algorithms such as **PPO, DPO and GRPO**; familiar with the **RAG** technology stack (Naive/Light/Graph), capable of designing and developing **Agentic Workflows**.

English: CET-6; strong ability to read English literature and write academically; follows daily SOTA papers on arXiv, and served as a main author of English technical reports and patents.